

Akash Uppala

Hyderabad, Telangana | uppalaakash2004@gmail.com | 8688965399 | linkedin.com/in/akash-uppala |
github.com/akash-uppala

PROFESSIONAL SUMMARY

Generative AI Engineer with 1+ year of production experience building LLM-powered systems, multimodal embedding pipelines, and vector search applications. Skilled in RAG architectures, agentic AI frameworks, and ML-based inference APIs. Experienced in deploying AI services end-to-end with CI/CD pipelines on GCP and AWS. Passionate about building explainable, scalable AI products that solve real-world problems.

EXPERIENCE

Software Engineer (AI/ML) | Mitt Arv Technologies Jul 2025 – Present

- Built an AI-powered phishing risk scoring platform using ML-driven feature extraction and real-time inference APIs, enabling explainable URL safety analysis for 10,000+ daily requests
- Developed multimodal embedding pipelines for text and image intelligence using vector search on MongoDB Atlas, powering semantic retrieval and similarity-based workflows with sub-100ms latency
- Built production-grade payment orchestration flows with idempotent retry handling, webhook reconciliation, and failure-safe order state transitions, reducing payment failures by ~35%
- Designed CI-backed test architecture and deployment pipelines for AI services on GCP, improving release confidence, regression coverage.

Software Engineer Intern | Mitt Arv Technologies Jan 2025 – Jun 2025

- Implemented authentication safety layer reducing unauthorized access and increasing daily active user engagement
- Built phone number verification system strengthening user authenticity, contributing to higher active user participation
- Redesigned core workflows into modular architecture, reducing codebase complexity by 30% and improving system performance and maintainability

PROJECTS

AutoAgent — LLM Reasoning Agent with Tool Use | LangGraph, OpenAI GPT-4o, FastAPI, Redis, Tavily Search 2025

- Built a multi-step LLM agent using LangGraph capable of autonomous web search, Python code execution, and API calls to complete complex research and analysis tasks end-to-end
- Implemented persistent agent memory using Redis enabling context retention across sessions, reducing redundant queries by 60%
- Integrated Tavily Search API as a real-time tool, enabling the agent to retrieve up-to-date information and synthesize structured reports with cited sources
- Tracked token usage, cost-per-run, and latency via custom observability layer; deployed as REST API on FastAPI with async task queuing

FineTuneKit — Domain-Specific Fine-Tuned LLM for Cybersecurity | Llama 3, QLoRA, Hugging Face, Unsloth, Weights & Biases, Python 2025

- Fine-tuned Llama 3 (8B) on a curated cybersecurity and phishing detection dataset using QLoRA with 4-bit quantization on free Kaggle GPU, achieving 92% classification accuracy on domain benchmarks
- Applied LoRA adapters to reduce trainable parameters by 99% vs full fine-tuning, enabling efficient domain adaptation with minimal compute cost
- Evaluated model performance using BLEU, ROUGE, and custom phishing-domain benchmarks; tracked experiments with Weights & Biases for reproducibility
- Published fine-tuned model and LoRA weights to Hugging Face Hub; built FastAPI inference endpoint for real-time cybersecurity Q&A and scam classification

TECHNICAL SKILLS

LLMs & GenAI: LangChain, LangGraph, OpenAI API (GPT-4o), Hugging Face, RAG Pipelines, Prompt Engineering, LLM Fine-Tuning (QLoRA/LoRA), Agentic AI, Llama 3, Mistral

ML & AI Tools: Sentence Transformers, Scikit-learn, Pandas, Matplotlib, Weights & Biases, Unsloth, FAISS, ChromaDB

Vector Databases: MongoDB Atlas Vector Search, ChromaDB, FAISS

Languages: Python, JavaScript, TypeScript, C++, Node.js

Frameworks & APIs: FastAPI, React, Celery, REST APIs, Streamlit

Cloud & DevOps: GCP, AWS (EC2, S3, RDS), Docker, GitHub Actions, CI/CD Pipelines

Developer Tools: Git, GitHub, VSCode, Cursor, Claude Code, Postman

Databases: MongoDB, MySQL

EDUCATION

Malla Reddy Engineering College Hyderabad, Telangana

B.Tech in Computer Science (Data Science) | CGPA: 7.9/10 Nov 2022 – May 2025

Government Institute of Electronics Hyderabad, Telangana

Diploma in Instrumentation Engineering | CGPA: 9.29/10 Jun 2019 – May 2022

RELEVANT COURSEWORK

Data Structures & Algorithms, Object-Oriented Systems, Operating Systems, SDLC, Machine Learning, Natural Language Processing